

LYAKHSH, Tajikistan

WMO: 387440

Lat: 38.317N

Lon: 71.267E

Elev: 5528

StdP: 11.99

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.0	2.3	-9.5	4.0	-0.6	-5.3	5.1	3.4	16.5	25.1	13.7	26.3	0.6	N/A	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.3	85.9	58.9	83.1	57.5	80.6	56.7	60.9	80.0	59.4	78.4	58.2	76.7	5.3	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.5	77.8	66.7	52.3	71.7	64.9	50.5	67.0	63.5	29.7	80.1	28.6	78.5	27.7	76.9	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	13.2	10.3	DB	-6.5	90.6	6.9	2.7	-11.5	92.5	-15.5	94.1	-19.3	95.6	-24.3	97.5	(4)
(5)			WB	-7.4	64.2	6.4	1.6	-12.1	65.4	-15.8	66.4	-19.4	67.3	-24.1	68.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.0	18.2	21.7	33.9	47.9	55.2	61.0	66.6	66.4	59.7	48.2	35.8	24.1	(6)	
	DBStd	18.09	8.89	8.68	8.76	6.62	4.57	4.45	4.64	4.56	5.12	5.75	6.86	6.94	(7)	
	HDD50	3743	986	794	501	116	9	1	0	0	3	104	426	803	(8)	
	HDD65	7483	1451	1214	965	514	305	133	36	34	166	522	875	1268	(9)	
	CDD50	1915	0	0	1	51	169	329	515	508	295	47	0	0	(10)	
	CDD65	182	0	0	0	0	0	11	86	77	8	0	0	0	(11)	
	CDH74	3039	0	0	0	4	35	284	1201	1210	300	5	0	0	(12)	
	CDH80	686	0	0	0	0	1	27	322	306	30	0	0	0	(13)	
(14) Wind	WSAvg	2.6	1.2	1.4	1.8	3.0	3.0	2.9	3.2	3.6	3.5	2.7	2.0	2.3	(14)	
(15) Precipitation	PrecAvg	20.7	2.0	2.5	3.0	2.7	2.3	1.3	1.1	0.6	0.4	1.2	1.7	2.0	(15)	
	PrecMax	25.9	4.1	4.3	4.7	4.3	4.7	2.4	2.8	1.1	0.7	3.1	4.4	3.8	(16)	
	PrecMin	15.3	0.9	1.0	1.1	1.1	1.2	0.6	0.4	0.3	0.0	0.4	0.6	0.7	(17)	
	PrecStd	2.9	0.8	0.7	0.9	0.8	0.9	0.5	0.5	0.2	0.2	0.6	0.9	0.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.9	46.5	59.9	73.3	78.1	83.3	89.9	89.8	83.3	74.0	60.9	43.2	(19)	
		MCWB	30.9	36.7	44.3	51.1	55.4	58.5	60.5	59.4	55.9	50.2	43.7	34.2	(20)	
	2%	DB	34.0	38.6	54.6	68.0	74.0	79.8	86.5	86.0	80.3	68.9	55.2	39.1	(21)	
		MCWB	29.7	33.6	42.4	49.5	53.5	56.8	59.5	58.5	54.3	48.4	40.9	32.9	(22)	
	5%	DB	32.1	35.2	49.7	64.3	70.9	77.4	83.5	83.2	77.5	65.0	50.9	36.3	(23)	
		MCWB	28.4	31.5	40.8	47.6	52.3	55.9	58.4	57.0	53.2	46.7	38.9	30.9	(24)	
	10%	DB	30.1	33.1	45.5	60.4	67.8	74.5	80.7	80.7	74.7	61.3	46.8	33.8	(25)	
		MCWB	26.9	30.0	38.7	46.6	51.1	55.2	57.6	56.6	52.0	45.0	37.4	29.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	37.8	45.9	52.9	56.6	60.2	63.5	62.9	58.2	51.5	44.6	36.4	(27)	
		MCDB	34.6	44.7	56.3	69.5	73.1	77.7	82.5	81.0	79.8	70.2	58.6	40.5	(28)	
	2%	WB	30.8	34.1	43.1	50.4	54.7	58.4	61.3	60.7	55.7	49.4	42.2	33.9	(29)	
		MCDB	33.1	38.0	52.6	64.4	70.4	75.5	80.5	79.6	76.5	65.8	52.2	37.8	(30)	
	5%	WB	28.8	32.0	41.2	48.7	53.3	57.2	59.9	59.0	54.0	47.8	40.3	31.7	(31)	
		MCDB	31.3	34.6	49.1	62.1	67.5	73.8	79.1	79.1	74.9	62.3	48.6	35.3	(32)	
	10%	WB	27.0	30.3	39.1	47.2	52.0	55.9	58.7	57.6	52.5	46.1	38.2	29.8	(33)	
		MCDB	30.0	32.9	45.4	58.8	65.1	71.8	77.2	77.1	72.2	58.4	45.4	33.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.6	14.8	18.4	21.3	24.3	26.3	27.8	27.5	23.0	16.9	13.8	(35)	
		MCDBR	13.7	13.5	20.5	26.2	28.0	29.0	30.4	30.7	31.5	29.0	24.2	15.9	(36)	
		MCWBR	10.5	9.9	10.7	11.4	10.9	10.5	9.8	10.4	11.5	12.2	12.7	11.1	(37)	
	5% WB	MCDBR	11.1	12.3	18.4	23.3	24.7	26.3	27.6	27.4	29.7	26.0	20.9	13.4	(38)	
		MCWBR	8.9	9.1	10.1	10.5	10.2	10.3	9.7	9.8	11.6	11.3	11.1	10.0	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.328	0.373	0.449	0.476	0.483	0.489	0.481	0.433	0.384	0.395	0.367	0.318	(40)	
	taud		2.036	1.915	1.800	1.834	1.874	1.879	1.923	2.067	2.170	2.048	2.060	2.124	(41)	
	Ebn at Noon		259	261	252	255	256	254	255	265	272	252	242	254	(42)	
	Edh at Noon		39	50	63	64	63	63	60	50	43	43	38	33	(43)	
(44) All-Sky Solar Radiation	RadAvg		798	1061	1497	1818	1963	2214	2304	2242	1879	1198	773	689	(44)	
	RadStd		56	97	108	158	150	142	121	112	95	145	113	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page